

4. Verify that the conversion is successful: `volume show vol_name -fields -volume-style -extended, state`

```
cluster-1::*> volume show my_volume -fields volume-style-extended, state
vserver volume      state  volume-style-extended
-----
vs0      my_volume online flexgroup
```

Results

The FlexVol volume is converted to a single-member FlexGroup volume.

After you finish

You can expand the FlexGroup volume, as required.

Convert a FlexVol volume SnapMirror relationship to a FlexGroup volume SnapMirror relationship

To convert a FlexVol volume SnapMirror relationship to a FlexGroup volume SnapMirror relationship in ONTAP, you must first convert the destination FlexVol volume followed by the source FlexVol volume.

What you'll need

- The FlexVol volume that is being converted must be online.
- The source FlexVol volume in the SnapMirror relationship must not be the source volume for multiple SnapMirror relationships.

Beginning with ONTAP 9.9.1, fanout SnapMirror relationships are supported for FlexGroup volumes. For more information, see [Considerations for creating SnapMirror cascade and fanout relationships for FlexGroups](#).

- The operations and configurations on the FlexVol volume must be compatible with the conversion process.

An error message is generated if the FlexVol volume has any incompatibility and the volume conversion is aborted. You can take corrective actions and retry the conversion.

About this task

FlexGroup conversion is supported only for asynchronous SnapMirror relationships.

Steps

1. Verify that the SnapMirror relationship is healthy: `snapmirror show`

Only XDP type mirror relationships can be converted.

```
cluster2::> snapmirror show
```

Progress	Source	Destination	Mirror	Relationship	Total		
Last	Path	Type	Path	State	Status	Progress	Healthy
Updated							
-----	-----	-----	-----	-----	-----	-----	-----
-----	vs0:src_dpv	DP	vs2:dst_dpv	Snapmirrored			
				Idle		-	true -
	vs0:src_xdp	XDP	vs2:dst_xdp	Snapmirrored			
				Idle		-	true -

2. Verify whether the source volume is compatible for conversion:

- Log in to the advance privilege mode: `set -privilege advanced`
- Verify the conversion process: `volume conversion start -vserver vs1 -volume src_vol -check-only true`

You must rectify all errors before converting the volume.

3. Convert the destination FlexVol volume to FlexGroup volume.

- Quiesce the FlexVol SnapMirror relationship: `snapmirror quiesce -destination-path dest_svm:dest_volume`

```
cluster2::> snapmirror quiesce -destination-path vs2:dst_xdp
```

- Start the conversion: `volume conversion start -vserver dest_svm -volume dest_volume`

```
cluster-1::> volume conversion start -vserver vs2 -volume dst_xdp
```

Warning: After the volume is converted to a FlexGroup, it will not be possible to change it back to a flexible volume.
Do you want to continue? {y|n}: y

[Job 510] Job succeeded: SnapMirror destination volume "dst_xdp" has been successfully converted to a FlexGroup volume.
You must now convert the relationship's source volume, "vs0:src_xdp", to a FlexGroup.
Then, re-establish the SnapMirror relationship using the "snapmirror resync" command.

4. Convert the source FlexVol volume to FlexGroup volume: `volume conversion start -vserver src_svm_name -volume src_vol_name`

```
cluster-1::> volume conversion start -vserver vs0 -volume src_xdp

Warning: Converting flexible volume "src_xdp" in Vserver "vs0" to a
FlexGroup
        will cause the state of all Snapshot copies from the volume to
be set
        to "pre-conversion". Pre-conversion Snapshot copies cannot be
        restored.
Do you want to continue? {y|n}: y
[Job 57] Job succeeded: success
```

5. Resync the relationship: `snapmirror resync -destination-path dest_svm_name:dest_volume`

```
cluster2::> snapmirror resync -destination-path vs2:dst_xdp
```

After you finish

You must ensure that when the source FlexGroup volume is expanded to include more constituents, the destination volume is also expanded.

Improve performance for multiple clients with FlexCache

You can use FlexCache volumes to speed up access to data or to offload traffic from heavily accessed volumes. FlexCache volumes are ideal for read-intensive workloads, especially where clients need to access the same data repeatedly.

The topics in this section show you how to manage FlexCache volumes with System Manager in ONTAP 9.7 and later releases. If you are using the classic System Manager (available only in ONTAP 9.7 and earlier), see this topic:

- [Create FlexCache volumes](#)

The FlexCache volume can be on the same cluster as or on a different cluster than that of the remote volume. If the remote volume is on a different cluster, you need to have already peered the clusters and storage VMs.



If an HA pair is using [encrypting SAS or NVMe drives \(SED, NSE, FIPS\)](#), you must follow the instructions in the topic [Returning a FIPS drive or SED to unprotected mode](#) for all drives within the HA pair prior to initializing the system (boot options 4 or 9). Failure to do this may result in future data loss if the drives are repurposed.

Steps

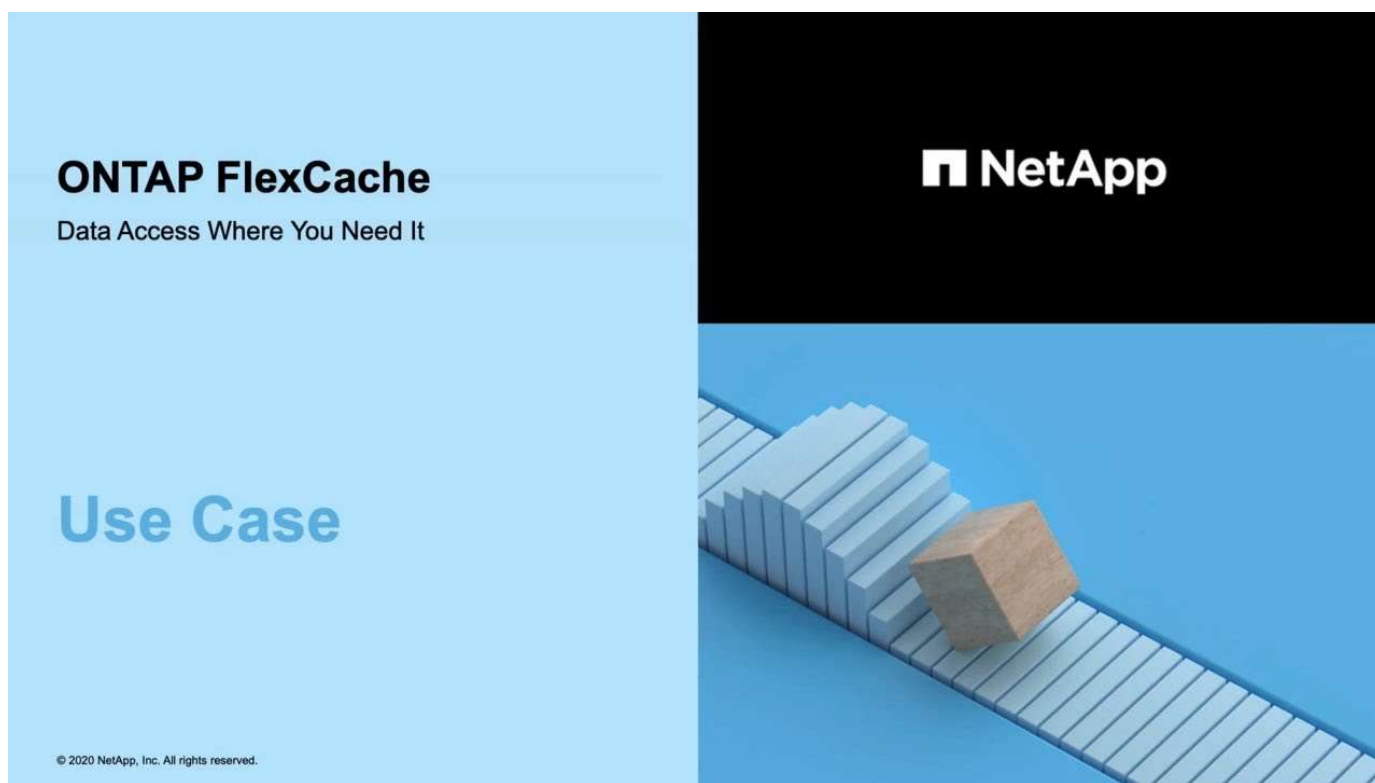
1. Click **Storage > Volumes**.
2. Click **Add**.

3. Click **More Options** and then select **Add as cache for a remote volume**.
 - a. If you are running ONTAP 9.8 or later and you want to disable QoS or choose a custom QoS policy, click **More Options** and then, under **Storage and Optimization** select **Performance Service Level**.

For any new data requests, the FlexCache volume requests the data from the remote volume and stores it. All the subsequent read requests for the data are then served directly from the FlexCache volume.

Videos

How FlexCache can reduce WAN latency and read times for global data



Learn about the performance benefits of ONTAP FlexCache!

ONTAP FlexCache

Data Access Where You Need It

Tech Clip

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 NetApp



FlexCache volumes management with the CLI

FlexCache volumes management overview with the CLI

You can configure and manage FlexCache volumes for accelerating data access.

Use these procedures to configure FlexCache volumes if the following are true:

- You are running ONTAP 9.5 or later.
- You want to use the ONTAP command-line interface (CLI), not System Manager or an automated scripting tool.
- You want to use best practices, not explore every available option.

Details about command syntax are available from the CLI help and the ONTAP man pages.

- You have cluster administrator privileges, not SVM administrator privileges.

Use FlexCache volumes to accelerate data access overview

A FlexCache volume is a sparsely populated volume that is backed by an origin volume. The FlexCache volume can be on the same cluster as or on a different cluster than that of the origin volume. The FlexCache volume provides access to data in the origin volume without requiring that all of the data be in the FlexCache volume.

In ONTAP 9.5, the origin volume is a FlexVol volume and the FlexCache volume is a FlexGroup volume. An origin volume supports NFSv3, NFSv4, and SMB protocols. A FlexCache volume supports only NFSv3 protocol in ONTAP 9.5. Beginning with ONTAP 9.8, a FlexCache volume also supports SMB protocol. Beginning with ONTAP 9.10.1, a FlexCache volume supports the NFSv4 protocol. For a table summary of

supported features in FlexCache volumes, refer to [Supported and unsupported features for FlexCache volumes](#).

Beginning with ONTAP 9.7, FlexGroup volumes are also supported as source volumes.



In ONTAP 9 releases earlier than 9.5, origin FlexVol volumes can only serve data to FlexCache volumes created on systems running Data ONTAP 8.2.x operating in 7-Mode. Beginning with ONTAP 9.5, origin FlexVol volumes can also serve data to FlexCache volumes on ONTAP 9 systems. For information about migrating from 7-mode FlexCache to ONTAP 9 FlexCache, [NetApp Technical Report 4743: FlexCache Volumes in NetApp ONTAP](#).

A FlexCache volume directly serves read requests if the volume contains the data requested by the client. Otherwise, the FlexCache volume requests the data from the origin volume and stores the data before serving the client request. Subsequent read requests for the data are then served directly from the FlexCache volume. This improves performance when the same data is accessed repeatedly, because after the first request, the data no longer has to travel across the network, or be served from an overloaded system.

Beginning with ONTAP 9.9.1, FlexCache volumes cache a directory listing for "file not found" errors that occur when a file no longer exists on the origin volume. This helps reduce network traffic by removing the need to send multiple calls to the origin when clients search for non-existent files.

Beginning with ONTAP 9.10.1, global file locking can be enabled across FlexCache volumes to favor consistency, ensuring modifications to an origin volume are distributed simultaneously to FlexCache volumes. Global file locking can only be enabled from the CLI.

You can use FlexCache volumes to speed up access to data or to offload traffic from heavily accessed volumes. FlexCache volumes help improve performance, especially when clients need to access the same data repeatedly, because the data can be served directly without having to access the origin volume. Therefore, you can use FlexCache volumes to handle system workloads that are read-intensive.

Any write operation is applied at the origin volume.

Typical FlexCache deployments

FlexCache volumes are typically used for read-intensive workloads. You can have a FlexCache volume in the same cluster to accelerate performance for frequently accessed data or "hot objects". You can also have FlexCache volumes to distribute data across multiple clusters to reduce WAN latencies.

You can have FlexCache deployments with AFF, FAS, or ONTAP Select systems. Beginning with ONTAP 9.6, FlexCache deployments are also supported with Cloud Volumes ONTAP.

Performance acceleration for hot volumes

In a LAN deployment, the FlexCache volume is in the same cluster as the origin cluster. The FlexCache volume can be located in the same SVM as or in a different SVM than that of the origin volume.

The FlexCache volume is used for CPU-intensive workloads to offload work from busy file servers and to free system resources. You can use multiple mount points corresponding to different FlexCache volumes for reducing network latency because the data access load is shared among all of the caching systems. This type of LAN deployment reduces the workload of an overloaded storage system.

Cross-cluster data distribution

In a WAN deployment, the FlexCache volume is remote from the data center and is in a different cluster than the origin volume. When clients request data, the FlexCache volume caches popular data, giving the end user faster access to information. This type of WAN deployment decreases the average access time for remote clients.

The FlexCache volume is placed as close as possible to the remote office. Client requests are then explicitly directed to the FlexCache volume. If valid data exists in the cache, that data is served directly to the client. If the data does not exist in the cache, the data is retrieved across the WAN from the origin system, cached in the FlexCache volume, and then served to the client.

Supported and unsupported features for FlexCache volumes

You must be aware of the features that are supported by FlexCache volumes and their origin volumes.

Feature	Supported at the origin volume?	Supported at the FlexCache volume?
Anti-ransomware protection	Yes Supported for FlexVol origin volumes beginning with ONTAP 9.10.1, not supported for FlexGroup origin volumes.	No
Antivirus	Yes Supported beginning with ONTAP 9.7	Not applicable
Auditing	Yes Supported beginning with ONTAP 9.7 You can audit NFS file access events in FlexCache relationships using native ONTAP auditing. For more information, see Considerations for auditing FlexCache volumes	Yes Supported beginning with ONTAP 9.7 You can audit NFS file access events in FlexCache relationships using native ONTAP auditing. For more information, see Considerations for auditing FlexCache volumes
Cloud Volumes ONTAP	Yes Supported beginning with ONTAP 9.6	Yes Supported beginning with ONTAP 9.6
Compaction	Yes Supported beginning with ONTAP 9.6	Yes Supported beginning with ONTAP 9.7

Compression	Yes Supported beginning with ONTAP 9.6	Yes Supported beginning with ONTAP 9.6
Deduplication	Yes	Yes Inline deduplication is supported on FlexCache volumes beginning with ONTAP 9.6. Cross-volume deduplication is supported on FlexCache volumes beginning with ONTAP 9.7.
FabricPool	Yes	Yes Supported beginning with ONTAP 9.7
FlexCache DR	Yes	Yes Supported beginning with ONTAP 9.9.1, with NFSv3 protocol, only. FlexCache volumes must be in separate SVMs or in separate clusters.
FlexGroup volume	Yes Supported beginning with ONTAP 9.7	Yes
FlexVol volume	Yes	No
FPolicy	Yes Supported beginning with ONTAP 9.7	Yes Supported for NFS beginning with ONTAP 9.7
MetroCluster configuration	Yes Supported beginning with ONTAP 9.7	Yes Supported beginning with ONTAP 9.7
Microsoft Offloaded Data Transfer (ODX)	No	No

NetApp Aggregate Encryption (NAE)	Yes Supported beginning with ONTAP 9.6	Yes Supported beginning with ONTAP 9.6
NetApp Volume Encryption (NVE)	Yes Supported beginning with ONTAP 9.6	Yes Supported beginning with ONTAP 9.6
NFSv3	Yes	Yes
NFSv4	Yes	Yes Supported beginning with ONTAP 9.10.1
QoS	Yes	Yes NOTE: File-level QoS is not supported for FlexCache volumes.
Qtrees	Yes Supported beginning with ONTAP 9.6	No
Quotas	Yes	No  Beginning with ONTAP 9.6, remote quota (rquota) is supported at FlexCache volumes.
SMB	Yes	Yes Supported beginning with ONTAP 9.8.
SMB Change Notify	Yes	No
SnapLock volumes	No	No

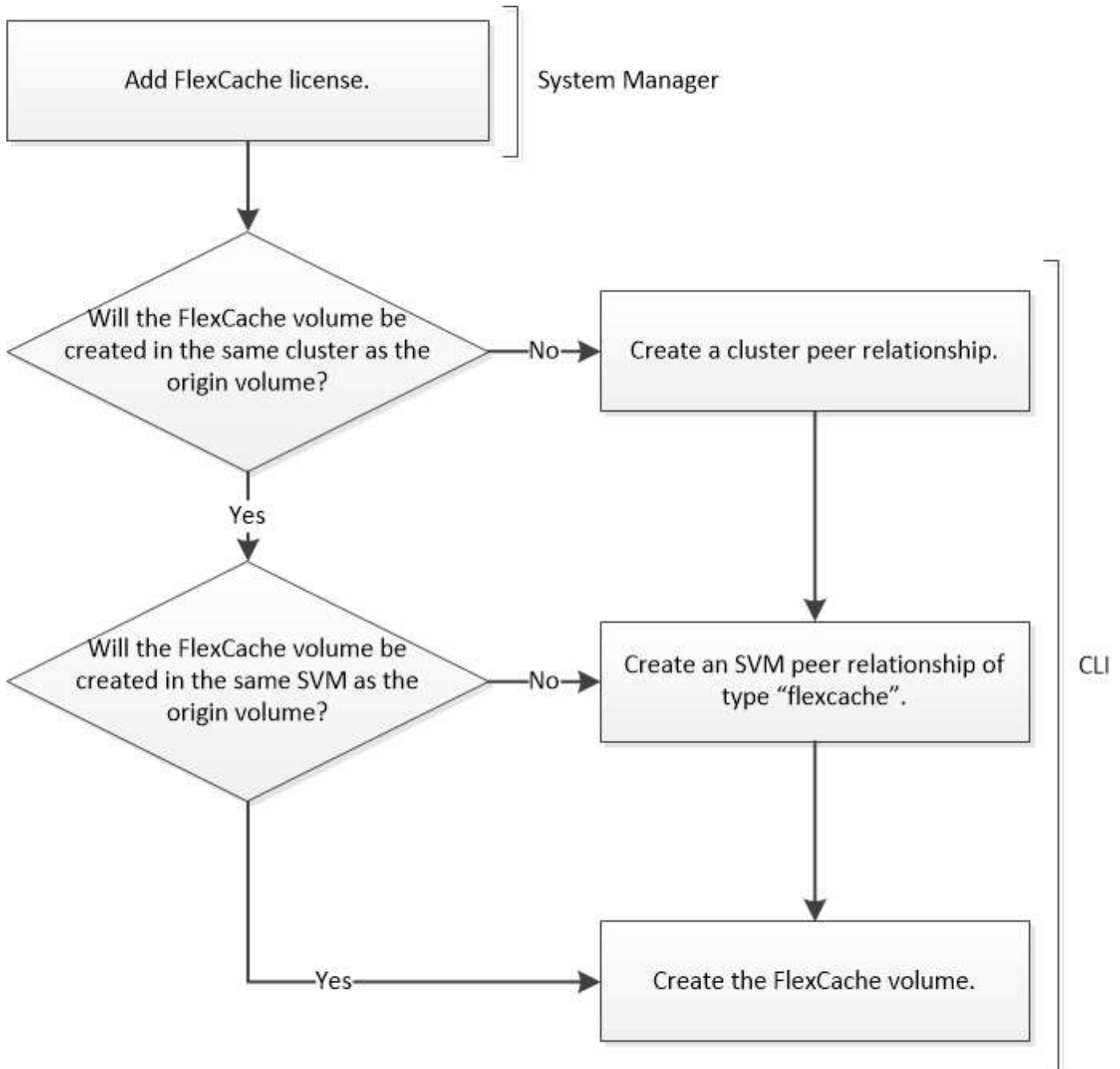
SnapMirror Asynchronous relationships	Yes	No - You can have a FlexCache volume from an origin primary volume in SnapMirror relationship. - Beginning with ONTAP 9.8, a SnapMirror secondary volume can be a FlexCache origin volume.
SnapMirror Synchronous relationships	No	No
SnapRestore	Yes	No
Snapshot copies	Yes	No
SVM DR configuration	Yes Supported beginning with ONTAP 9.5. The primary SVM of an SVM DR relationship can have the origin volume; however, if the SVM DR relationship is broken, the FlexCache relationship must be re-created with a new origin volume.	No You can have FlexCache volumes in primary SVMs, but not in secondary SVMs. Any FlexCache volume in the primary SVM is not replicated as part of the SVM DR relationship.
Storage-level Access Guard (SLAG)	No	No
Thin provisioning	Yes	Yes Supported beginning with ONTAP 9.7
Volume cloning	Yes Cloning of an origin volume and the files in the origin volume is supported beginning with ONTAP 9.6.	No
Volume move	Yes	Yes (only for volume constituents) Moving volume constituents of a FlexCache volume is supported from ONTAP 9.6 onwards.

Volume rehost	No	No
---------------	----	----

FlexCache volume creation

FlexCache volume creation workflow

You must first install the FlexCache license from System Manager. You can then create a FlexCache volume in the same cluster or in a remote cluster by using the CLI.



You must be running ONTAP 9.5 or later.

You can use FlexCache volumes in the same cluster for accelerated performance when accessing hot volumes. You can use FlexCache volumes in different clusters for improving the performance of cross-cluster

data distribution.

Add a FlexCache license

If you are running ONTAP 9.6 or earlier, you must install a FlexCache license, which is a capacity-based and term-based license, by using System Manager.

About this task

The FlexCache license is a cluster-wide license. The license includes an entitled usage limit that you purchase for using FlexCache volumes in the cluster. The space usage by FlexCache volumes across the cluster must not exceed the capacity of the entitled usage limit. If you need to increase the usage limit of the license, you should contact your sales representative.

Beginning with ONTAP 9.7, the capacity-based license is not required. The FlexCache license is bundled with the ONTAP package.

Steps

1. Download the NetApp License File for the FlexCache license from the NetApp Support Site.

[NetApp Support](#)

2. Use System Manager to upload the FlexCache license to the cluster:
 - a. Click the **Configurations > Cluster > Licenses** tab.
 - b. In the **Packages** window, click **Add**.
 - c. In the **Add License Packages** dialog box, click **Choose Files** to select the NetApp License File that you downloaded, and then click **Add** to upload the file to the cluster.

Process to create a FlexCache volume

Create a FlexCache volume

You can create a FlexCache volume in the same cluster for improving performance when accessing a hot object. If you have data centers in different locations, you can create FlexCache volumes on remote clusters for accelerating data access.

About this task

The FlexCache volume is always a FlexGroup volume, and not a FlexVol volume.

Beginning with ONTAP 9.7, FlexGroup volumes are also supported at the origin of the FlexCache relationship.

Steps

1. If the FlexCache volume to be created is in a different cluster, create a cluster peer relationship:
 - a. On the destination cluster, create a peer relationship with the data protection source cluster:

```
cluster peer create -generate-passphrase -offer-expiration MM/DD/YYYY  
HH:MM:SS|1...7days|1...168hours -peer-addr peer_LIF_IPs -initial-allowed  
-vserver-peers svm_name,...* -ip-space ip-space_name
```

Beginning with ONTAP 9.6, TLS encryption is enabled by default when creating a cluster peer relationship. TLS encryption is supported for the intercluster communication between the origin and FlexCache volumes. You can also disable TLS encryption for the cluster peer relationship, if required.

```
cluster02::> cluster peer create -generate-passphrase -offer
-expiration 2days -initial-allowed-vserver-peers *
```

Passphrase: UCa+6lRVICXeL/gq1WrK7ShR
Expiration Time: 6/7/2017 08:16:10 EST
Initial Allowed Vserver Peers: *
Intercluster LIF IP: 192.140.112.101
Peer Cluster Name: Clus_7ShR (temporary generated)

Warning: make a note of the passphrase - it cannot be displayed again.

- b. On the source cluster, authenticate the source cluster to the destination cluster:

```
cluster peer create -peer-addr peer_LIF_IPs -ip space ip space
```

```
cluster01::> cluster peer create -peer-addr
192.140.112.101,192.140.112.102
```

Notice: Use a generated passphrase or choose a passphrase of 8 or more characters.

To ensure the authenticity of the peering relationship, use a phrase or sequence of characters that would be hard to guess.

Enter the passphrase:
Confirm the passphrase:

Clusters cluster02 and cluster01 are peered.

2. If the FlexCache volume is in a different SVM than that of the origin volume, create an SVM peer relationship with flexcache as the application:

- a. If the SVM is in a different cluster, create an SVM permission for the peering SVMs:

```
vserver peer permission create -peer-cluster cluster_name -vserver svm-name
-applications flexcache
```

The following example illustrates how to create an SVM peer permission that applies for all of the local SVMs:

```
cluster1::> vsserver peer permission create -peer-cluster cluster2
-vserver "*" -applications flexcache
```

Warning: This Vserver peer permission applies to all local Vservers. After that no explicit "vsserver peer accept" command required for Vserver peer relationship creation request from peer cluster "cluster2" with any of the local Vservers. Do you want to continue? {y|n}: y

b. Create the SVM peer relationship:

```
vsserver peer create -vserver local_SVM -peer-vserver remote_SVM -peer
-cluster cluster_name -applications flexcache
```

3. Create a FlexCache volume:

```
volume flexcache create -vserver cache_svm -volume cache_vol_name -auto
-provision-as flexgroup -size vol_size -origin-vserver origin_svm -origin
-volume origin_vol_name
```

The following example creates a FlexCache volume and automatically selects existing aggregates for provisioning:

```
cluster1::> volume flexcache create -vserver vs_1 -volume fc1 -auto
-provision-as flexgroup -origin-volume vol_1 -size 160MB -origin-vserver
vs_1
[Job 443] Job succeeded: Successful
```

The following example creates a FlexCache volume and sets the junction path:

```
cluster1::> flexcache create -vserver vs34 -volume fc4 -aggr-list
aggr34,aggr43 -origin-volume origin1 -size 400m -junction-path /fc4
[Job 903] Job succeeded: Successful
```

4. Verify the FlexCache relationship from the FlexCache volume and the origin volume.

a. View the FlexCache relationship in the cluster:

```
volume flexcache show
```

```
cluster1::> volume flexcache show
Vserver Volume      Size      Origin-Vserver Origin-Volume Origin-
Cluster
-----
vs_1      fc1        160MB     vs_1          vol_1
cluster1
```

- b. View all of the FlexCache relationships in the origin cluster:

```
volume flexcache origin show-caches
```

```
cluster::> volume flexcache origin show-caches
Origin-Vserver Origin-Volume  Cache-Vserver  Cache-Volume  Cache-
Cluster
-----
vs0             ovol1          vs1            cfg1           clusA
vs0             ovol1          vs2            cfg2           clusB
vs_1            vol_1          vs_1           fc1
cluster1
```

Result

The FlexCache volume is successfully created. Clients can mount the volume by using the junction path of the FlexCache volume.

Related information

[Cluster and SVM peering](#)

[ONTAP 9 Commands](#)

Guidelines for sizing a FlexCache volume

You must be aware of the limits for FlexCache volumes before you start provisioning the volumes.

The size limit of a FlexVol volume is applicable to an origin volume. The size of a FlexCache volume can be less than or equal to the origin volume. The best practice for the size of a FlexCache volume is to be at least 10 percent of the size of the origin volume.

You must also be aware of the following additional limits on FlexCache volumes:

Limit	ONTAP 9.5-9.6	ONTAP 9.7	ONTAP 9.8 and later

Maximum number of FlexCache volumes that you can create from an origin volume	10	10	100
Recommended maximum number of origin volumes per node	10	100	100
Recommended maximum number of FlexCache volumes per node	10	100	100
Recommended maximum number of FlexGroup constituents in a FlexCache volume per node	40	800	800
Maximum number of constituents per FlexCache volume per node	32	32	32

Related information

[NetApp Interoperability](#)

Considerations for auditing FlexCache volumes

Beginning with ONTAP 9.7, you can audit NFS file access events in FlexCache relationships using native ONTAP auditing and file policy management with FPolicy. FPolicy is not supported for FlexCache volumes with SMB. Native auditing and FPolicy are configured and managed with the same CLI commands used for FlexVol volumes. However, there is some different behavior with FlexCache volumes.

• Native auditing

- You can't use a FlexCache volume as the destination for audit logs.
- If you want to audit read and writes on FlexCache volumes, you must configure auditing on both the cache SVM as well as on the origin SVM.

This is because file system operations are audited where they are processed. That is, reads are audited on the cache SVM and writes are audited on the origin SVM.

- To track the origin of write operations, the SVM UUID and MSID are appended in the audit log to identify the FlexCache volume from which the write originated.
- Although system access control lists (SACLs) can be set on a file using NFSv4 or SMB protocols, FlexCache volumes support only NFSv3. Therefore, SACLs can only be set on the origin volume.

• FPolicy

- Although writes to a FlexCache volume are committed on the origin volume, FPolicy configurations monitor the writes on the cache volume. This is unlike native auditing, in which the writes are audited on the origin volume.
- While ONTAP does not require the same FPolicy configuration on cache and origin SVMs, it is recommended that you deploy two similar configurations. You can do so by creating a new FPolicy policy for the cache, configured like that of the origin SVM but with the scope of the new policy limited to the cache SVM.

Manage a FlexCache relationship

View the connection status of a FlexCache relationship

Beginning with ONTAP 9.6, you can view the connection status of a FlexCache relationship and take any corrective action if the connection status between the origin and FlexCache volumes goes to the disconnected mode.

About this task

A FlexCache relationship can have one of the following connection status:

- connected
- disconnected
- unknown

Steps

1. Log in to the advanced privilege mode:

```
set -privilege advanced
```

2. Verify the connection status of all the FlexCache relationships in the cluster:

```
volume flexcache connection-status show
```

```
cluster::*> volume flexcache connection-status show
```

```
Node: cluster-01
```

		Remote		Remote
Connection	Volume	Vserver	Remote Volume	Endpoint
+Vserver Status				
+-----	-----	-----	-----	-----
vs_1 connected	vol_origin	vs_2	fc_11__0001	cache
vs_1 connected	vol_origin	vs_2	fc_11__0002	cache
vs_1 connected	vol_origin	vs_2	fc_11__0003	cache
vs_1 connected	vol_origin	vs_2	fc_11__0004	cache
vs_2 connected	fc_11	vs_1	vol_origin	origin

Synchronize properties of a FlexCache volume from an origin volume

Some of the volume properties of the FlexCache volume must always be synchronized with those of the origin volume. If the volume properties of a FlexCache volume fail to synchronize automatically after the properties are modified at the origin volume, you can manually synchronize the properties.

About this task

The following volume properties of a FlexCache volume must always be synchronized with those of the origin volume:

- Security style (-security-style)
- Volume name (-volume-name)
- Maximum directory size (-maxdir-size)
- Minimum read ahead (-min-readahead)

Step

1. From the FlexCache volume, synchronize the volume properties:

```
volume flexcache sync-properties -vserver svm_name -volume flexcache_volume
```

```
cluster1::> volume flexcache sync-properties -vserver vs1 -volume fcl
```

Update the configurations of a FlexCache relationship

After events such as volume move, aggregate relocation, or storage failover, the volume configuration information on the origin volume and FlexCache volume is updated automatically. In case the automatic updates fail, an EMS message is generated and then you must manually update the configuration for the FlexCache relationship.

If the origin volume and the FlexCache volume are in the disconnected mode, you might need to perform some additional operations to update a FlexCache relationship manually.

About this task

If you want to update the configurations of a FlexCache volume, you must run the command from the origin volume. If you want to update the configurations of an origin volume, you must run the command from the FlexCache volume.

Step

1. Update the configuration of the FlexCache relationship:

```
volume flexcache config-refresh -peer-vserver peer_svm -peer-volume  
peer_volume_to_update -peer-endpoint-type [origin | cache]
```

Enable file access time updates

Beginning with ONTAP 9.11.1, you can enable the `-atime-update` field on the FlexCache volume to permit file access time updates. You can also set an access time update period with the `-atime-update-period` attribute. The `-atime-update-period` attribute controls how often access time updates can take place and when they can propagate to the origin volume.

Overview

ONTAP provides a volume-level field called `-atime-update`, to manage access time updates on files and directories that are read using `READ`, `READLINK`, and `READDIR`. Atime is used for data lifecycle decisions for files and directories that are infrequently accessed. The infrequently accessed files are eventually migrated to archive storage and are often later moved to tape.

The `atime-update` field is disabled by default on existing and newly created FlexCache volumes. If you are using FlexCache volumes with ONTAP releases earlier than 9.11.1, you should leave the `atime-update` field disabled so caches aren't unnecessarily evicted when a read operation is performed on the origin volume. With large FlexCache caches, however, administrators use special tools to manage data and help to ensure that hot data remains in the cache and cold data is purged. This is not possible when `atime-update` is disabled.

However, beginning with ONTAP 9.11.1, you can enable `-atime-update` and `-atime-update-period`, and use the tools required to manage the cached data.

Before you begin

All FlexCache volumes must be running ONTAP 9.11.1 or later.

About this task

Setting `-atime-update-period` to 86400 seconds allows no more than one access time update per 24-hour period, regardless of the number of read-like operations performed on a file.

Setting the `-atime-update-period` to 0 sends messages to the origin for each read access. The origin then informs each FlexCache volume that the atime is outdated, which impacts performance.

Steps

1. Enable file access time updates and set the update frequency:

```
volume modify -volume vol_name -vserver SVM_name -atime-update true -atime-update-period seconds
```

The following example enables `-atime-update` and sets `-atime-update-period` to 86400 seconds, or 24 hours:

```
c1: volume modify -volume origin1 vs1_c1 -atime-update true -atime-update-period 86400
```

2. Verify that `-atime-update` is enabled:

```
volume show -volume vol_name -fields atime-update,atime-update-period
```

```
c1::*> volume show -volume cache1_origin1 -fields atime-update,atime-update-period
vserver volume          atime-update atime-update-period
-----
vs2_c1  cache1_origin1 true           86400
```

Enable global file locking

Beginning with ONTAP 9.10.1, global file locking can be applied to prevent reads across all related cached files.

About this task

By default, FlexCache volumes favor availability over consistency. Without global file locking, any modification to an origin will be distributed to FlexCache volumes, but they might not be updated simultaneously. Global file locking favors consistency across volumes over availability. With global file locking enabled, modifications to the origin will be suspended until all FlexCache volumes are online.